KABALE UNIVERSITY

BCS 3101: BASICS OF MACHINE LEARNING

DATA PRE-PROCESSING AND EXPLORATORY DATA ANALYSIS

Agricultural Crop Yield in Indian States Dataset

Assignment 2 Report

Student Name: Kyoshabire Dianah

Registration Number: 2024/A/5188/G/F

1. Background

Agriculture is an important source of food, employment and income. Crop yield is affected by several interacting factors, which may include the crop type, season, geographical location, cultivated area, rainfall and the use of agricultural inputs. Understanding the relationships among these variables can help researchers and agricultural planners examine patterns in crop production and identify information that may be useful when developing predictive models.

This project uses the Agricultural Crop Yield in Indian States Dataset, selected from the Agriculture domain in the BCS 3101 project options. The intended machine-learning problem is a supervised regression task: using suitable information in the dataset to estimate the crop-yield target. The analysis begins with inspecting and preparing the data, followed by exploratory data analysis (EDA). The purpose of this report is to document the dataset, explain the preprocessing decisions, and interpret the patterns revealed by the actual notebook outputs.

The report is a working draft. Values in square brackets must be replaced with the results produced by the student's own Jupyter notebook. No missing-value counts, outlier counts, post-cleaning row counts, charts or model results have been assumed.

2. Data Description

2.1 Dataset source

Dataset: Agricultural Crop Yield in Indian States Dataset. Source: Kaggle, uploaded by akshatgupta7. Dataset page: https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset

The course project brief lists the dataset as having approximately 19,689 rows and 10 columns. These are catalogue figures; the exact dimensions used in this analysis must be confirmed from the downloaded CSV using df.shape.

3. Problem Statement and Objectives

3.1 Problem statement

Crop-yield information is influenced by a combination of agricultural, seasonal and location-related factors. When these variables are stored in raw datasets, missing values, duplicate records, unusual observations, inconsistent data types and differences in measurement scales can make analysis difficult. A structured preprocessing and exploratory analysis process is therefore needed before the data can be used reliably for machine learning.

This project investigates how the selected agricultural dataset can be inspected, cleaned and prepared for a regression task that estimates crop yield from relevant available features. The findings may be useful to students, agricultural researchers and planners interested in understanding historical crop-production patterns. The dataset represents Indian states, so conclusions should not automatically be treated as representative of Uganda or other countries.

3.2 General objective

To prepare and explore the Agricultural Crop Yield in Indian States Dataset so that it is suitable for subsequent machine-learning analysis of crop yield.

3.3 Specific objectives

1. To inspect the structure, variables, data types and dimensions of the selected agricultural dataset.
2. To identify and appropriately handle missing values, duplicate records and data-quality problems.
3. To detect and assess potential outliers in relevant numerical variables.
4. To apply suitable data transformation and reduction/preparation techniques, including scaling where appropriate.
5. To use visualisations and descriptive statistics to identify patterns and relationships relevant to the crop-yield target.

4. Data Preprocessing Techniques Applied

Preprocessing should be carried out in a documented sequence. The notebook is the evidence for each operation. The following subsections describe the methods to report; complete the bracketed fields using the actual outputs from the notebook.

4.1 Initial inspection

The CSV file was loaded into a Pandas DataFrame named df. The first records were inspected with df.head(), the dimensions with df.shape, and the column names with df.columns. The df.info() and df.describe() outputs were used to understand data types and numerical summaries.

4.2 Missing-value assessment and treatment

Missing values were checked using df.isnull().sum(). Missing values can reduce the number of usable records or cause errors in some machine-learning algorithms. The treatment should depend on the amount of missing data and the meaning of the affected variable. Numerical variables may be imputed using an appropriate statistic such as the median, while categorical variables may use the mode. Rows should only be removed when justified.

4.3 Duplicate-record assessment

Duplicate records were checked using df.duplicated().sum(). Exact accidental duplicates can over-represent repeated observations and should be removed where appropriate. A record should not be removed merely because two genuine agricultural observations happen to share some values.

4.4 Outlier detection and treatment

Potential outliers in numerical variables were examined using boxplots and, where appropriate, the interquartile range (IQR) rule. Q1 is the 25th percentile and Q3 is the 75th percentile. IQR = Q3 − Q1. Values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are flagged for investigation. A flagged value is not automatically an error: unusually high crop production or yield may be a genuine observation. Removal, capping or retention should be justified by the data and project context.

4.5 Data transformation and scaling

Data transformation prepares variables in a form that is more suitable for analysis. Categorical variables used as predictors may require encoding, while numerical variables may require scaling.

StandardScaler transforms a numerical feature by subtracting its training-set mean and dividing by its training-set standard deviation. The transformed training feature therefore has a mean near zero and standard deviation near one.

Scaling is not required for every model. It is often useful for scale-sensitive algorithms, but tree-based models generally do not require standardisation. To avoid data leakage, the data should be split first; the scaler should be fitted only on the training predictors and then used to transform the test predictors.

4.6 Data reduction and train/test preparation

Data reduction means simplifying the dataset while retaining information relevant to the task. This may involve removing irrelevant identifiers, selecting useful features, or reducing dimensions when justified. Any removed column should be explained. For a regression task, the target variable must be separated from the predictor matrix before model training.

The dataset should be divided into training and test sets, for example using an 80:20 split and a fixed random_state for reproducibility. The test set must remain separate until final evaluation.

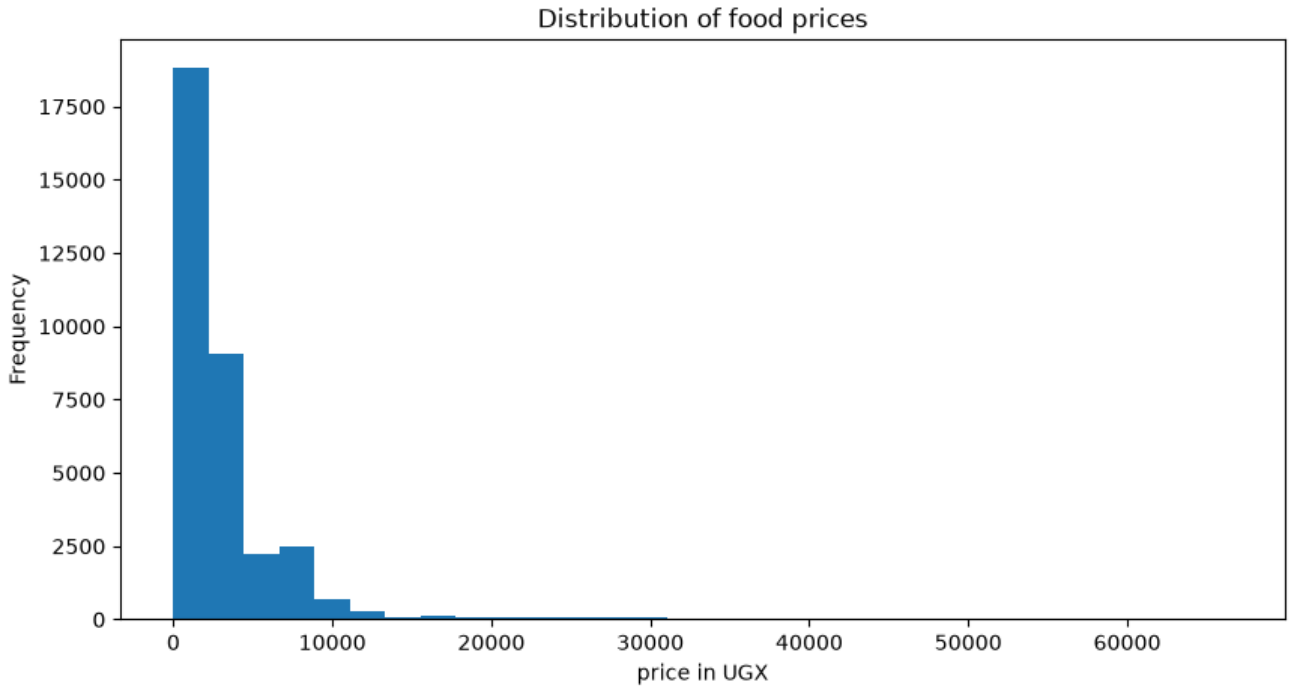
Split used: [e.g., 80% training and 20% testing].

5. Exploratory Data Analysis (EDA)

EDA uses descriptive statistics and visualisations to understand the data before modelling. Each figure inserted in the final report should have a figure number, a descriptive caption and a short interpretation based on the actual chart.

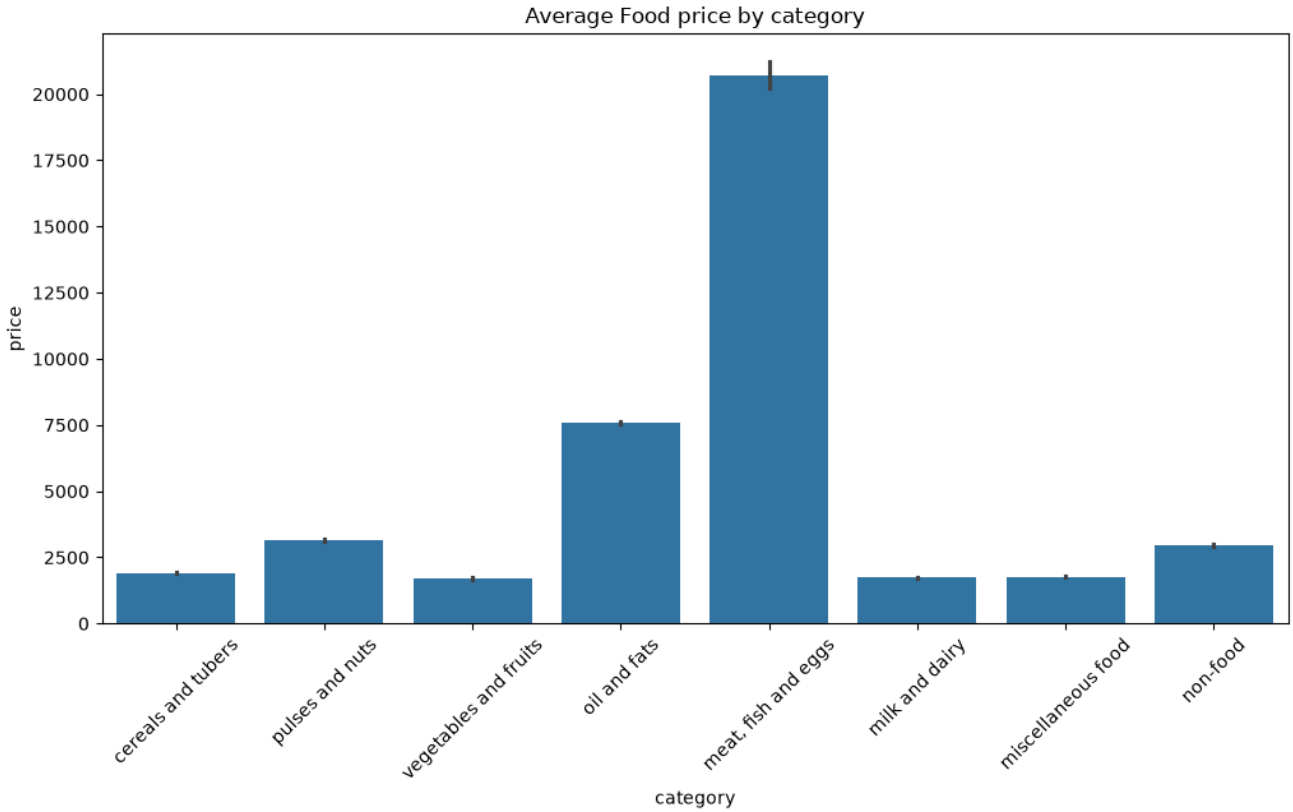
5.1 Descriptive statistics

The df.describe() summary was used to examine the count, mean, standard deviation, minimum, quartiles and maximum of numerical variables. These statistics help identify differences in scale, skewness and possible extreme values.



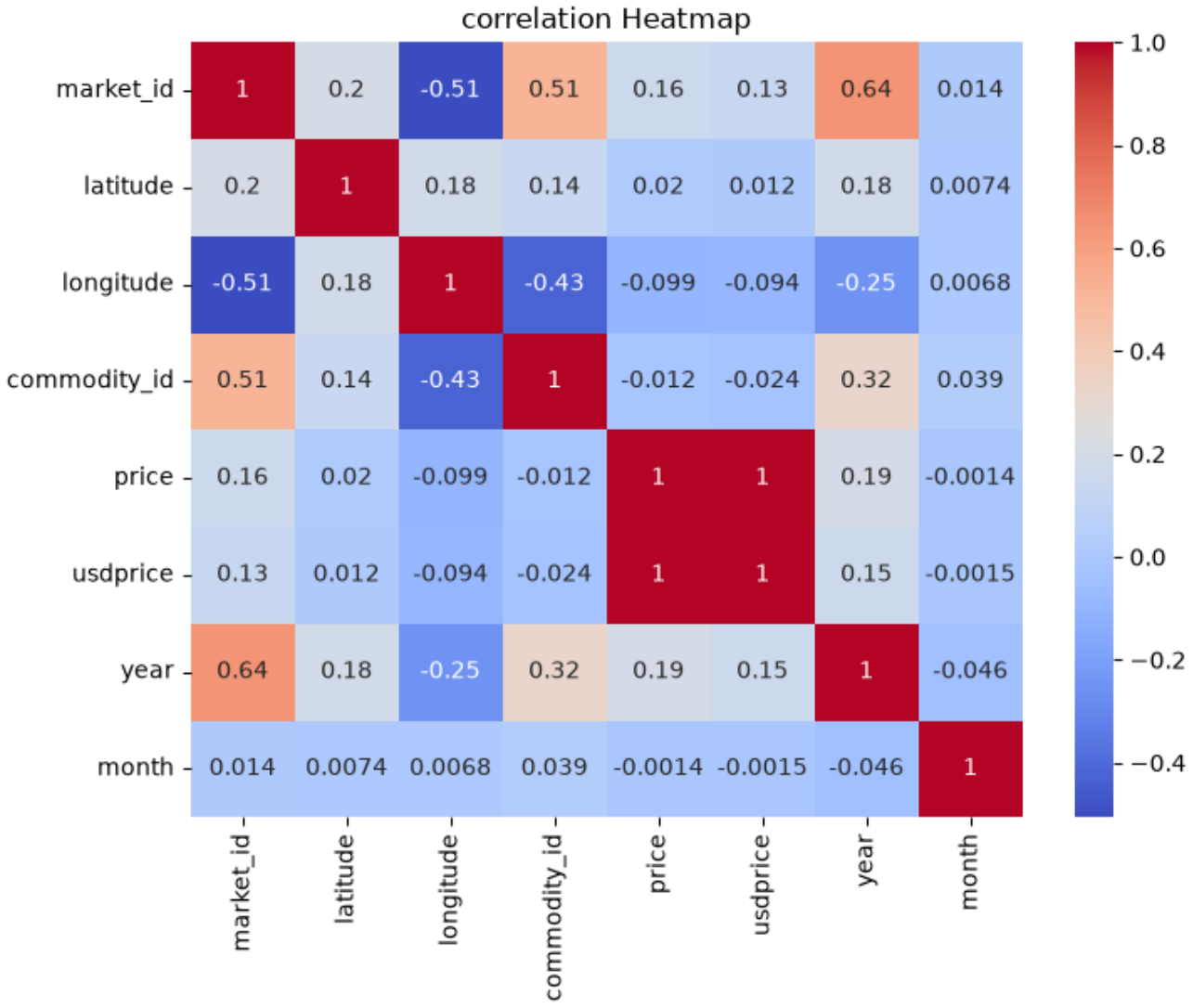
5.2 Distribution of the target variable

A histogram or density plot of the selected target variable should be used to examine its distribution. The plot can show whether values are concentrated in a particular range or whether the distribution is strongly skewed. If the target is skewed, this may influence the choice of evaluation metrics or a possible target transformation.



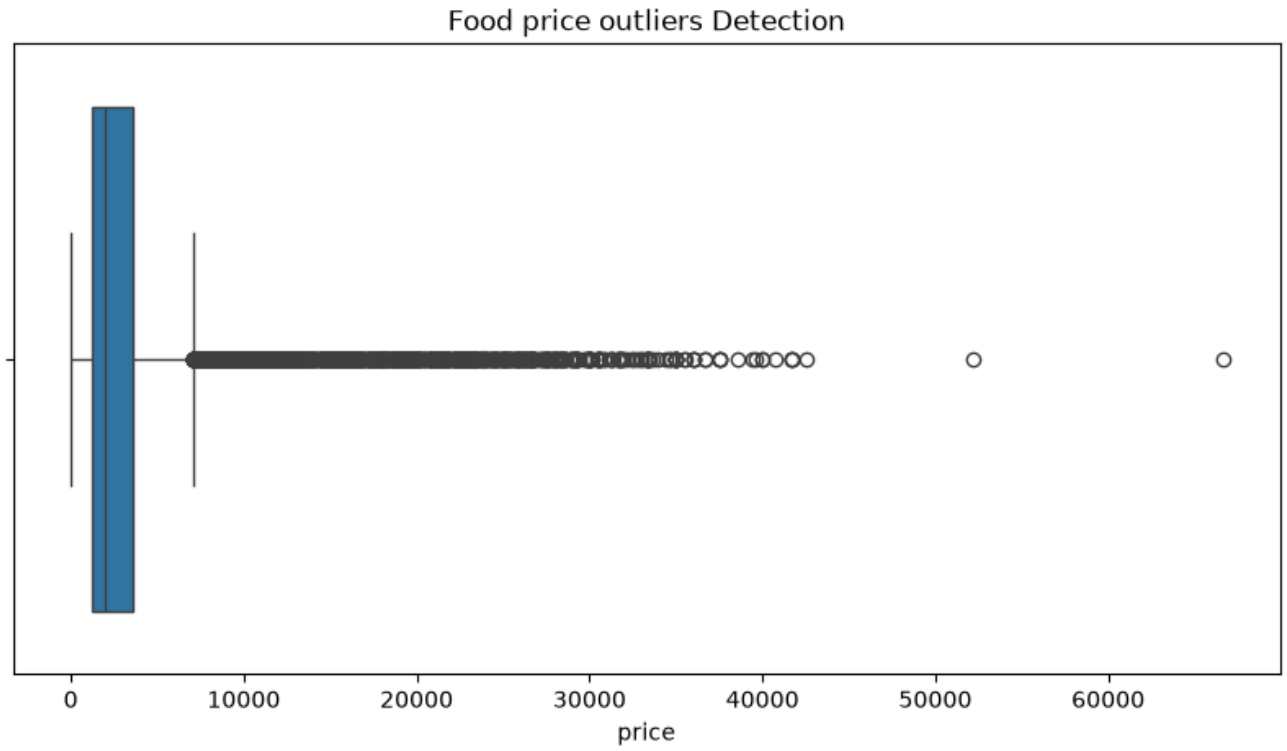
5.3 Relationships among numerical variables

A correlation matrix or heatmap can show the direction and strength of linear relationships among numerical variables. Correlation does not prove causation. It can, however, help identify variables that may be informative and pairs of predictors that are strongly related.



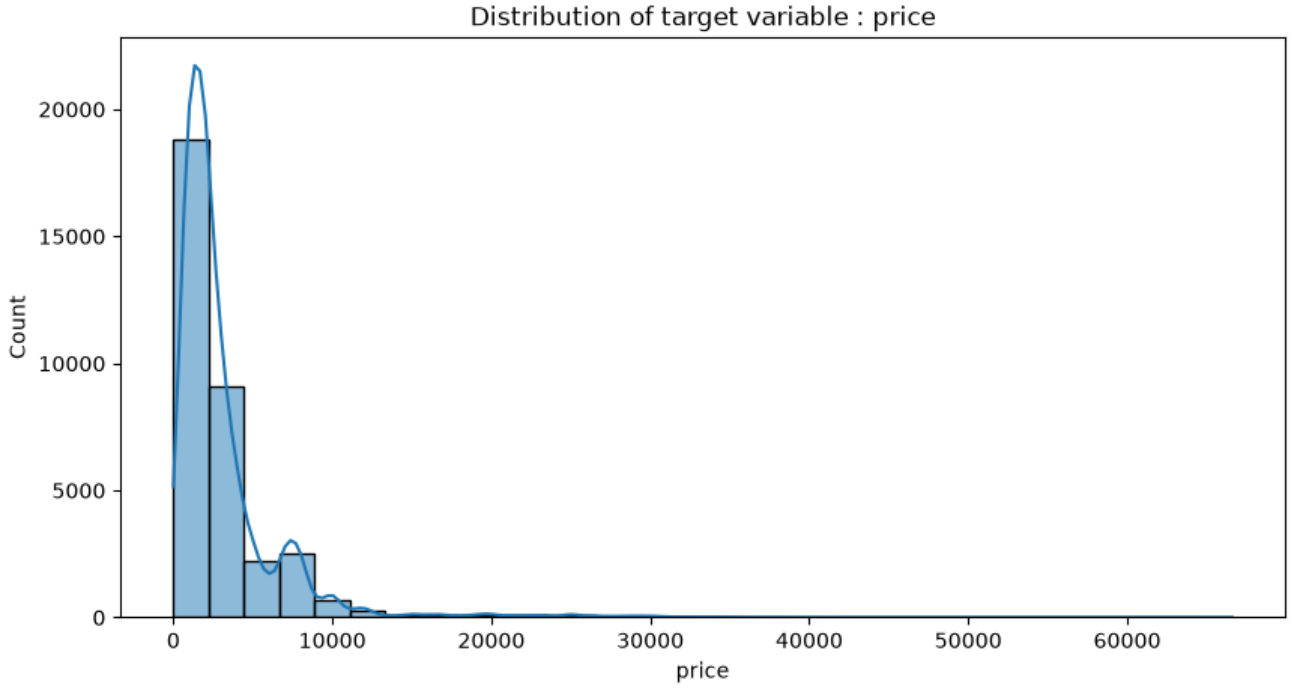
5.4 Outlier visualisation

Boxplots were used to visualise the spread of numerical features and highlight observations outside the whiskers. These points were treated as potential outliers for investigation, not automatically as incorrect data.



5.5 Comparison across categories (if applicable)

If the dataset contains categorical variables such as crop, season or state, grouped plots can compare the target across categories. Use only categories that actually appear in df.columns and ensure labels are readable.



6. Conclusion and Next Steps

This report has described the initial preparation and exploratory analysis of the Agricultural Crop Yield in Indian States Dataset. The work examined the dataset structure and documented the treatment of missing values, duplicates, potential outliers, transformations and preparation for a training/test split. The EDA section is intended to show the distribution of the target and the relationships among relevant variables.

Based on the completed notebook, the dataset's readiness for modelling should be summarised here: [state whether the data is ready and identify any remaining limitations]. The next step is to train and compare suitable regression models, such as Linear Regression, Decision Tree Regressor and Random Forest Regressor, using appropriate evaluation measures such as MAE, RMSE and R-squared. Model performance must be reported only after the models have actually been trained and tested.

7. References

6. akshatgupta7. (n.d.). Agricultural Crop Yield in Indian States Dataset. Kaggle. https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset
7. Kabale University. (n.d.). BCS 3101: Basics of Machine Learning — Assignment 2 Companion Guide: The Data Pre-processing & Exploratory Data Analysis Pipeline. [Course material].
8. Kabale University. (n.d.). BCS 3101: Basics of Machine Learning — Practice Project. [Course material].
9. Tukey, J. W. (1977). Exploratory data analysis. Addison-Wesley.